ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Bloch's Theorem, Multi-Sublattice Transformations & 1D Chains

I. Crystal Translational Symmetry & The Bloch Theorem

II. Constructing the Tight-Binding Bloch Sum

III. Monatomic 1D Chain

IV. Crystals with a Basis: The Multi-Sublattice Problem

I. Crystal Translational Symmetry & The Bloch Theorem

Review:

Lattice Translation Vector: $\boldsymbol{R_n = n_1 a_1 + n_2 a_2 + n_3 a_3}$**, where,** $\boldsymbol{a_i}$ **are primitive vectors**

Translation Operator ($\boldsymbol{T_R}$**): Defines the spatial displacement of a wavefunction by a lattice vector** $\boldsymbol{R}$

$$\boldsymbol{T_R}\Psi(\boldsymbol{r}) = \Psi(\boldsymbol{r} + \boldsymbol{R})$$

Periodic Potential: The crystal potential satisfies $\boldsymbol{V(r - R_n) = V(r)}$**, which implies the Hamiltonian commutes with the translation operator:** $[\boldsymbol{H}, \boldsymbol{T_n}] = \boldsymbol{0}$

Physical Consequence: According to quantum mechanics, commuting operators share a simultaneous set of eigenstates. Therefore, the energy stationary states $\Psi(\boldsymbol{r})$ **must also be unique eigenstates of the translation operator** $\boldsymbol{T_R}\Psi(\boldsymbol{r}) = \lambda_{\boldsymbol{R}}\Psi(\boldsymbol{r})$

I. Crystal Translational Symmetry & The Bloch Theorem

Boundary Condition: Consider a macroscopically large but finite crystal with N_i cells along each primitive direction a_i

Apply the periodic boundary condition $\Psi(\boldsymbol{r}+\boldsymbol{N_i a_i})=\Psi(\boldsymbol{r}) \Rightarrow \boldsymbol{T_{N_i a_i}}\Psi(\boldsymbol{r})=\lambda_{a_i}^{N_i}\Psi(\boldsymbol{r})=\Psi(\boldsymbol{r})$

Solving the Eigenvalue: This enforces $\lambda_{a_i}^{N_i}=1$**, which means** λ_{a_i} **must be a root of unity:** $\lambda_{a_i}=\boldsymbol{e}^{i2\pi m_i/N_i}$

where, $\boldsymbol{m_i}$ **is an integer**

Defining the Wavevector $\boldsymbol{k}$**: For a general lattice vector** $\boldsymbol{R}=\sum \boldsymbol{N_i a_i}$**, the accumulated eigenvalue is:** $\lambda_{\boldsymbol{R}}=\boldsymbol{e}^{i\boldsymbol{k}\cdot\boldsymbol{R}}$

where, $\mathbf{k}=\sum\frac{\boldsymbol{m_i}}{\boldsymbol{N_i}}\boldsymbol{b_i}$ **and** $\boldsymbol{b_i}$ **is the lattice vector of unit cell within the reciprocal space**

I. Crystal Translational Symmetry & The Bloch Theorem

The Redundancy Paradox: Let *G* be a reciprocal lattice vector satisfying $e^{iG\cdot R} = 1$

Shift the wavevector *k* by *G*, the translation eigenvalue remains completely unchanged

$$\lambda_R = e^{i(k+G)\cdot R} = e^{ik\cdot R} \cdot e^{iG\cdot R} = e^{ik\cdot R}$$

Bloch's Theorem: The eigenstates of a periodic system must satisfy $\Psi_k(r + R_n) = e^{ik\cdot R_n}\Psi_k(r)$

where, *k* is the crystal wavevector confined within the First Brillouin Zone (1st BZ)

Physical Meaning: *k* and *k* + *G* describe the exact same physical quantum state with identical translation symmetry

The First Brillouin Zone (1st BZ): To eliminate this mathematical redundancy, one restricts all unique *k* vectors to the unique Wigner-Seitz cell of the reciprocal lattice—the First Brillouin Zone

II. Constructing the Tight-Binding Bloch Sum

The Problem: Raw atomic orbitals $\Phi_\alpha(r - R_i)$ do not satisfy Bloch's theorem individually

The Solution (Bloch Sum): Linearly combine atomic orbitals across all lattice sites with a phase factor $e^{ik\cdot R_n}$

$$|\Psi_\alpha(r)\rangle = \frac{1}{\sqrt{N}}\sum_n e^{ik\cdot R_n}|\Phi_\alpha(r - R_i)\rangle$$ **where, *N* is the total number of unit cells**

Under the orthogonal Bloch sum basis, $|\Psi_\alpha(r)\rangle$, calculate the *k* dependent Hamiltonian matrix element

$$H_{\alpha\nu}(k) = \langle\Phi_\alpha(k)|H|\Phi_\nu(k)\rangle$$

$$\downarrow$$

$$H_{\alpha\nu}(k) = \frac{1}{N}\sum_{m,n} e^{-ik\cdot R_m}e^{ik\cdot R_n}\langle\Phi_\alpha(r - R_m)|H|\Phi_\nu(r - R_n)\rangle$$

II. Constructing the Tight-Binding Bloch Sum

Using lattice translational invariance, let $R_j = R_n - R_m$:

$$H_{\alpha\nu}(k) = \sum_j e^{ik\cdot R_j} t_{\alpha\nu}(R_j)$$

where, $t_{\alpha\nu}(R_j)$ is the real-space hopping integral

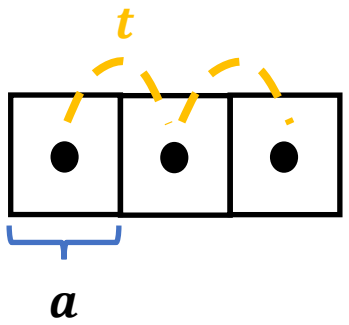
Standard format of the tight-binding Hamiltonian matrix

The Secular Equation: $det[H_k - E_k I] = 0$

III. Monatomic 1D Chain

System Geometry: A 1D array of identical atoms separated by lattice constant a

Real space lattice vectors $R_n = na$



Basis: One single *s* orbital per atom (α = 1)

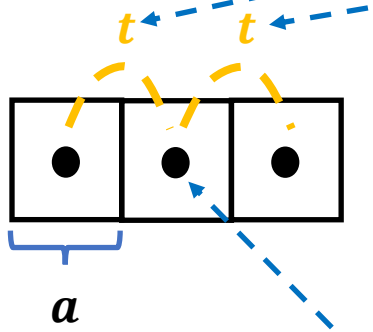
$H(k)$ will be a 1×1 matrix (a scalar)

TBM Parameters:

- **On-site energy: $\langle \Phi(r - na)|H|\Phi(r - na)\rangle = \varepsilon_0$**
- **Nearest-neighbor (*NN*) hopping: $\langle \Phi(r - na)|H|\Phi(r - (n \pm 1)a)\rangle = t$**
- **All further hoppings, ($2a$, $3a$, …), are truncated to 0**

III. Monatomic 1D Chain

Nearest-neighbor (*NN*) hopping: $\langle \Phi(\boldsymbol{r}-\boldsymbol{na})|\boldsymbol{H}|\Phi(\boldsymbol{r}-(\boldsymbol{n}\pm\boldsymbol{1})\boldsymbol{a})\rangle = \boldsymbol{t} \longrightarrow \boldsymbol{H_{is,js}} = \boldsymbol{te^{ik\cdot -a}} + \boldsymbol{te^{ik\cdot a}}$



On-site energy: $\langle \Phi(\boldsymbol{r}-\boldsymbol{na})|\boldsymbol{H}|\Phi(\boldsymbol{r}-\boldsymbol{na})\rangle = \boldsymbol{\varepsilon_0} \longrightarrow \boldsymbol{H_{is,is}} = |\boldsymbol{\varepsilon_0}|$

Total Hamiltonian matrix is $\boldsymbol{H} = \boldsymbol{\varepsilon_0} + \boldsymbol{t}\left(\boldsymbol{e^{ik\cdot -a}} + \boldsymbol{e^{ik\cdot a}}\right)$

III. Monatomic 1D Chain

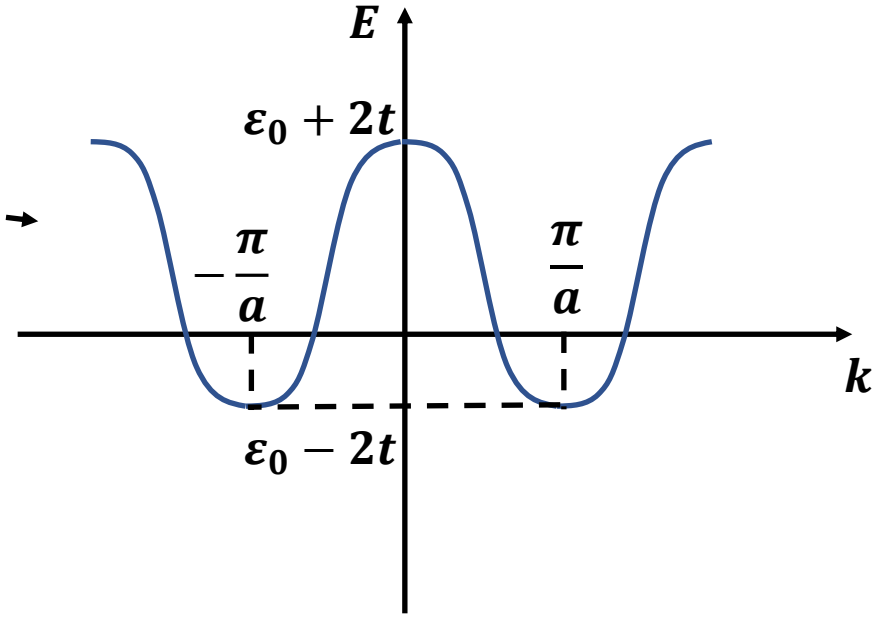
Solving the Hamiltonian and obtain the eigenstate and eigenvalue

$$det[H - EI] = det\left[\varepsilon_0 + t\left(e^{ik\cdot -a} + e^{ik\cdot a}\right) - EI\right] = 0$$

$$E = \varepsilon_0 + 2t \cdot cos(k \cdot a)$$

Boundary Conditions: *k* is confined within the First Brillouin Zone: $-\frac{\pi}{a} \leq k \leq \frac{\pi}{a}$

Band Structure



III. Monatomic 1D Chain

$$E = \varepsilon_0 + 2t \cdot cos(k \cdot a)$$



Bandwidth: $E_{max} - E_{min} = (\varepsilon_0 + 2t) - (\varepsilon_0 - 2t) = 4t$

Group Velocity: $v_g(k) = \frac{1}{\hbar}\frac{dE(k)}{dk} = -\frac{2ta}{\hbar} sin(k \cdot a)$

Effective Mass: $m^*(k) = \hbar^2 \left(\frac{d^2E(k)}{d^2k}\right)^{-1} = -\frac{\hbar^2}{2ta^2 cos(k \cdot a)}$

- **Bottom of band (k = 0):** $m^* = -\frac{\hbar^2}{2ta^2} > 0$ **(parabolic electron-like)**
- **Top of band (k = 0):** $m^* = \frac{\hbar^2}{2ta^2} < 0$ **(hole-like)**

IV. Crystals with a Basis: The Multi-Sublattice Problem

Bravais Lattice: A purely mathematical array of points where every single point has an identical geometric environment (*e.g.*, 1D monatomic chain, Simple Cubic)

Crystal Structure in reality: Bravais Lattice ⊗ Basis (Atomic Cluster)

Crystal Structure = Bravais Lattice + Basis

The Non-Bravais Reality: Many of the most vital materials in spintronics and valleytronics—such as Graphene (Hexagonal), Silicon (Diamond), and MoS_2—are NOT Bravais lattices

They consist of multiple interpenetrating Bravais sublattices

IV. Crystals with a Basis: The Multi-Sublattice Problem

In a monatomic crystal, every atom is located precisely at R_n. The translation vector connects all atoms and Bloch Sum applies for such case; while, in a crystal with a basis, atoms are located at $R_n + \tau_\mu$. The distance between atom A and atom B inside the same unit cell is $\tau_B - \tau_A$, which is NOT a lattice translation vector. Under such case, Bloch Sum does not work any longer

The Consequence

$$T_{R_n}\Psi(r-\tau_\mu) \neq \Psi(r-\tau_\mu)$$

One cannot use a single, global Fourier transform to diagonalize the Hamiltonian. If one forces them into a single Bloch sum, the matrix elements will lose their phase tracking, leading to mathematically incorrect band structures

IV. Crystals with a Basis: The Multi-Sublattice Problem

Solution:

Define the position of an atom as: $r_{n,\mu} = R_n + \tau_\mu$, where R_n is the unit cell vector, and τ_μ is the internal coordinate of sublattice μ (*e.g.* $\mu = A, B$)

Construct independent Bloch sums for each sublattice separately: $$|\Phi_{\alpha\nu}(k)\rangle = \frac{1}{\sqrt{N}}\sum_n e^{ik\cdot R_n}|\Phi_\alpha(r - R_n - \tau_\mu)\rangle$$

Crucial Definition Note: The phase factor inside the sum uses the unit cell position: R_n, NOT the exact atomic position $R_n + \tau_\mu$

This is the standard convention for multi-sublattice TBM

IV. Crystals with a Basis: The Multi-Sublattice Problem

Intra- *v.s.* Inter- Sublattice Matrix Elements

The generalized Hamiltonian matrix $H(k)$ now has a block structure index by sublattices (α, ν)

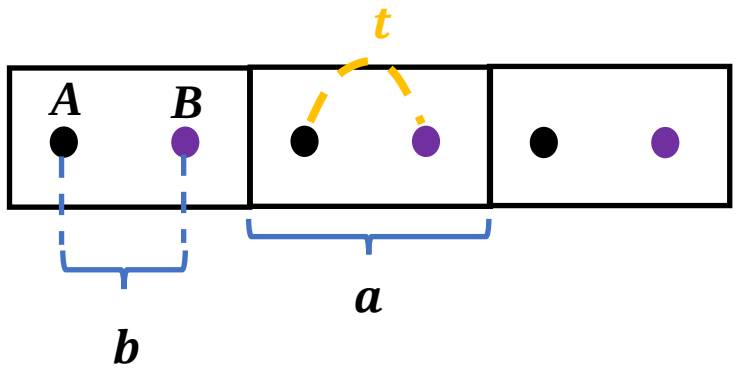
$$H_{\alpha\nu}(\boldsymbol{k}) = \sum_{j} e^{i\boldsymbol{k}\cdot\boldsymbol{R}_j} t_{\alpha\nu}(\boldsymbol{R}_j)$$

Diagonal Elements ($H_{\alpha\alpha}$): Represent Intra-sublattice hopping (hopping within the same sublattice) and on-site energies

Off-Diagonal Elements ($H_{\alpha\nu}$, $\alpha\neq\nu$): Represent Inter-sublattice hopping (hopping from sublattice α to sublattice ν)

IV. Crystals with a Basis: The Multi-Sublattice Problem

Definition: Only consider hopping between atoms that are immediate geometric neighbors. All $t_{\alpha\nu}$ for 2nd, 3rd and higher coordination shells are set to exactly zero



$$H(k) = \left(\sum_{\langle ii \rangle} \varepsilon_i + \sum_{\langle i,j \rangle} e^{ik\cdot b} t_{\alpha\nu} \right) + h.c. \xrightarrow{\text{Matrix Form}}$$

	A	B
A	ε_A	$e^{ik\cdot b}t$
B	$e^{ik\cdot -b}t$	ε_B

IV. Crystals with a Basis: The Multi-Sublattice Problem

$$det\begin{vmatrix} \varepsilon_A\text{-}E & e^{ik\cdot b}t \\ e^{ik\cdot -b}t & \varepsilon_B\text{-}E \end{vmatrix} = 0$$

$$E_{\pm}(k) = \frac{\varepsilon_A + \varepsilon_B}{2} \pm \sqrt{\left(\frac{\varepsilon_A - \varepsilon_B}{2}\right)^2 + t^2}$$

Band Gap: $|E_+(k) - E_-(k)| = 2\sqrt{\left(\frac{\varepsilon_A - \varepsilon_B}{2}\right)^2 + t^2}$



That's all for this session

Thank you for your attention and great participation